

Geralt Heart

Austin TX • geraldheart01@gmail.com • 956-802-1359 • github.com/GHeart01 • linkedin.com/in/geraldheart001 • geraldheart.com

SUMMARY

MS Computer Science student at CU Boulder with a focus on Machine Learning, Data mining, Network Systems, and Computer Vision. Software engineering experience in production environments building automation and internal tools with UDP networking layers. Work in Python, SQL, and full-stack deployment; driven to automate.

WORK EXPERIENCE

Software Engineering Intern | Athena Manufacturing LP – Austin, TX *May 2026 – Aug 2026*

- Built and deployed an internal Kanban-style automated scheduler pulling live data from Microsoft SQL Server, replacing a manual process for the machinist team; utilizing pyodbc, PyQt6, Pandas, and JSON
- Created a local persistence layer with per user authentication for database access and account management. Utilized pymssql to validate user identity when accessing M2M data and secure multi-user access for storing individual settings and preferences.
- Engineered a UDP-based networking layer to optimize database calls by up to 90% and designed a multi-recipient mode that simplified packet transfer from several packets into single transmissions.
- Developed automated scheduling algorithms to minimize retooling on machines and maximize throughput across several departments/work-centers.

AI Trainer | Handshake – Austin, TX

Feb 2026 – Present

- Evaluated AI-generated content for factual accuracy and monitored hallucinations. Supported Reinforcement Learning from Human Feedback pipelines to optimize performance across projects.
- Contributed to multi-project RLHF workflows between reasoning, coding, and general knowledge tasks.

PROJECTS

Regression Testing Framework • *python, pytest, GitHub Actions*

github.com/GHeart01/tile-gps-regression-testing

- Designed 52 automated unit and regression tests covering 4 system modules (TILE, EMQuest GPS, Spectra Python, and Aurora systems) achieving 100% test validation with execution time under 0.1 seconds.
- Validated Android GPS module functionality including location accuracy, signal strength, and RTK/DGPS compatibility using integration tests for data persistence, serialization, JSON validation, and configuration.
- Created CI/CD pipeline using GitHub Actions with automated testing, coverage analysis, & multi-version Python support.

Pricing Web Scraper • *PostgreSQL, Python, BeautifulSoup, REST API*

github.com/GHeart01/Web-Scraper-Clothing

- Implemented a PostgreSQL relational database schema, indexing the price of men's clothing across major retailers.
- Utilized rate limiting to manage anti-scraping protections (403 errors) and evaluated performance against REST API alternatives. Created master scraper controller that aggregates results from multiple sources, calculates price comparisons, and exports normalized JSON data compatible with PostgreSQL

Customer Segmentation & Clustering Analysis • *NumPy, Pandas, seaborn, matplotlib, TensorFlow*

github.com/GHeart01/Wholesale_Customer_Clustering

- Built a customer segmentation pipeline using K-Means and Hierarchical Clustering to identify purchasing behavior patterns. Applied hyperparameter tuning across several different models to evaluate performance.
- Evaluated clustering quality using silhouette scores between multiple cluster counts, provided heatmap/pair plot visualizations to demonstrate feature relationships.

EDUCATION

University of Colorado Boulder • GPA: 3.9/4.0

Jan 2025 – May 2027

Master of Science in Computer Science

Boulder, CO

University of Texas Rio Grande Valley • GPA: 3.5/4.0

Jul 2017 – May 2020

Bachelor of Science in Applied Mathematics

Edinburg, TX

SKILLS

Programming Languages: Python, SQL, C++, JavaScript, MATLAB, Bash, PowerShell

Frameworks & Libraries: PyQt6, React, Node.js, pytest, Three.js, Pandas, NumPy, scikit-learn, TensorFlow, Keras, SciPy

Databases: PostgreSQL, MySQL, MS SQL Server

Cloud & DevOps: AWS Lightsail, Docker, Kubernetes